# The FREUID Challenge 2026

## Technical Report — janetanand

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**Abstract**

*The FREUID Challenge 2026 (IJCAI-ECAI) requires classifying identity-document images as bona-fide or fraudulent, where attacks span GenAI-driven edits, physical print-and-capture workflows, and the held-out private test set contains document types unseen during training. Submissions are ranked by the official FREUID Score (lower is better), a harmonic mean of AuDET and APCER@1%BPCER. We present a transfer-learning pipeline built on EfficientNet-B2 with progressive backbone unfreezing, StratifiedGroupKFold cross-validation by document type, and a 5-fold ensemble. The solution reaches a public leaderboard FREUID Score of 0.271 and 0.27106 after private test inference.*

Competition: The FREUID Challenge 2026 (IJCAI-ECAI 2026)

Kaggle team: janetanand

Code: https://github.com/Janet-Priya/freuid-challenge-2026

# 1. Introduction

Identity document fraud detection is a security-critical task underpinning remote onboarding, financial services, and border control. The FREUID Challenge 2026, hosted at IJCAI-ECAI, targets a realistic threat model: attacks include GenAI-driven edits, physical tampering, and print-and-capture attacks where a forged document is printed and re-photographed — suppressing pixel-level artifacts and closing the so-called analog hole. Two properties make the challenge difficult. First, the attack space is heterogeneous, spanning global layout inconsistencies and fine-grained local artifacts. Second, the private test set contains two document types entirely absent from training, demanding genuine cross-domain generalization.

Our approach pairs EfficientNet-B2 transfer learning with a principled cross-validation strategy (StratifiedGroupKFold by document type) that directly simulates the unseen-type private test structure, a custom implementation of the competition's evaluation metric, and progressive backbone unfreezing. The solution is trained on a single NVIDIA GTX 1650 (4 GB VRAM) and packaged as an offline Docker container for reproducibility.

# 2. Method

## 2.1 Problem Formulation

The task is binary classification: given an identity-document image, emit a continuous fraud score P(fraud) $\in [0, 1]$, where higher values indicate greater confidence of fraud. The evaluation metric is the FREUID Score:

$$\textbf{FREUID} = 2 \times (\text{AuDET} \times \text{APCER@1\%BPCER}) / (\text{AuDET} + \text{APCER@1\%BPCER})$$

where AuDET is the area under the Detection Error Trade-off curve (lower is better), and APCER@1%BPCER is the Attack Presentation Classification Error Rate at a 1% Bona-Fide Presentation Classification Error Rate operating point (lower is better). The harmonic mean penalizes any weakness in either component.

## 2.2 Model Architecture

We use EfficientNet-B2 pretrained on ImageNet, loaded via the timm library, with its classification head removed (num_classes=0) to obtain a 1408-dimensional feature vector. A custom classifier head is attached:

- Linear(1408 → 256) → ReLU → Dropout(0.5) → Linear(256 → 1)

Progressive backbone unfreezing was applied iteratively across experiments. Starting from only the final block (blocks.6) and conv_head trainable, we incrementally unfroze blocks.5, blocks.4, and finally blocks.3, finding monotonic improvement in both cross-validation score and public leaderboard score. The type embedding (mapping document type to a learned 16-dimensional vector) used in early experiments was removed after finding it introduced train/test mismatch: test document types are unknown at inference, making the embedding a source of noise.

## 2.3 Validation Strategy

StratifiedGroupKFold (n_splits=5) grouped by the document type column ensures each validation fold contains a document type unseen during training. With 5 document types, this maps cleanly to 5 folds:

| Fold | Val Type | Train Types |
|---|---|---|
| 0 | MOZAMBIQUE/DL | EGYPT, GUINEA, BENIN, MAURITIUS |
| 1 | BENIN/DL | EGYPT, GUINEA, MOZAMBIQUE, MAURITIUS |
| 2 | GUINEA/DL | EGYPT, BENIN, MOZAMBIQUE, MAURITIUS |
| 3 | MAURITIUS/ID | EGYPT, GUINEA, BENIN, MOZAMBIQUE |
| 4 | EGYPT/DL | GUINEA, BENIN, MOZAMBIQUE, MAURITIUS |

This mirrors the private test's unseen-type structure. Standard random splits were avoided as they permit template memorization and produce over-optimistic validation scores.

## 2.4 Training Configuration

| Parameter | Value |
|---|---|
| Backbone | EfficientNet-B2 (pretrained ImageNet) |
| Image size | 320 × 320 |
| Batch size | 4 |
| Optimizer | Adam |
| Learning rate | 3e-5 |
| Weight decay | 1e-3 |
| Loss | BCEWithLogitsLoss |
| Early stopping | Patience 3 |
| Max epochs | 10 |
| Hardware | NVIDIA GTX 1650, 4 GB VRAM |

## 2.5 Data Augmentation

Training augmentations applied in sequence:

- RandomHorizontalFlip
- RandomRotation (±15°)
- ColorJitter (brightness 0.3, contrast 0.3, saturation 0.3, hue 0.05)
- RandomPerspective (distortion 0.4, p=0.7)
- GaussianBlur (kernel 5, sigma 0.1–3.0)

- RandomAffine (translate 0.1, scale 0.9–1.1)
- RandomErasing (p=0.3, scale 0.02–0.1)

A custom RecaptureSimulation augmentation chains perspective warp, affine transform, brightness/contrast shift, and Gaussian blur in sequence with p=0.6 to simulate the full print-and-photograph physical pipeline. A MoireNoise transform adding periodic interference patterns to the tensor was also tested but found to hurt at 320px resolution and was excluded from the final recipe.

# 3. Data

## 3.1 Training Set

| Metric | Value |
|---|---|
| Total training samples | 69,352 |
| Bona-fide (label 0) | 40,005 (57.7%) |
| Fraudulent (label 1) | 29,347 (42.3%) |
| Document types | 5 |
| Digital fraud (is_digital=True) | 69,332 (99.97%) |
| Recaptured fraud (is_digital=False) | 20 (0.03%) |

The training set covers five document types: EGYPT/DL, GUINEA/DL, BENIN/DL, MOZAMBIQUE/DL, MAURITIUS/ID. Notably, 99.97% of fraud samples are purely digital — recaptured samples are almost entirely absent from training, while the competition organizers noted that the private test set contains more recaptured documents.

## 3.2 External Data

No external datasets were used. All training data derives from the official FREUID training set.

## 3.3 Pre-trained Models

EfficientNet-B2 ImageNet pretrained weights via timm (Apache 2.0 license). No other pretrained models were used.

# 4. Inference

At inference, predictions from all 5 fold models are averaged (ensemble):

**Final score = mean(sigmoid(model_0(x)), ..., sigmoid(model_4(x)))**

No test-time augmentation was used in the final submission to minimize inference runtime and comply with the 6-hour A100 constraint. Inference is run on the public test set (7,821 images) and private test set (134,997 images) separately, with predictions combined into a single 142,818-row submission file.

Docker inference commands:

```
docker build -t freuid-repro:local . docker run --rm --network none --gpus all --shm-size=2g \   -v /path/to/test/images:/data:ro \   -v $(pwd)/out:/submissions freuid-repro:local
```

# 5. Results

## 5.1 Cross-Validation Progression

| Experiment | Mean CV FREUID |
|---|---|
| EfficientNet-B2, blocks.6 only | 0.222 |
| + Remove type embedding | 0.212 |
| + blocks.5+6 unfrozen, 320px | 0.191 |
| + blocks.4+5+6 unfrozen | 0.157 |
| + blocks.3+4+5+6 unfrozen | 0.153 |

### 5.2 Public Leaderboard Progression

| Submission | Public Score |
|---|---|
| EfficientNet-B2 + type embedding | 0.358 |
| No type embedding | 0.377 |
| 320px resolution | 0.329 |
| 320px + unfreeze blocks.5+6 | 0.313 |
| 320px + unfreeze blocks.4+5+6 | 0.274 |
| Full private+public inference | 0.271 ← best |

Progressive backbone unfreezing was the most impactful single change — each additional unfrozen block improved both CV and public leaderboard score consistently. Removing the type embedding improved generalization on hard folds (unseen document types). 320px resolution provided consistent gains over 224px.

## 6. Reproducibility

### 6.1 Environment

- Python 3.10
- PyTorch ≥ 2.0.0 (CUDA 12.1)
- timm ≥ 0.9.0
- scikit-learn ≥ 1.2.0
- Pillow, pandas, numpy, tqdm

### 6.2 Reproduction Steps

1. Install dependencies:

```
pip install -r requirements.txt
```

2. Place training images at ~/freuid/train/train/ and train_folds.csv at ~/freuid/

3. Train all 5 folds:

```
python train_unfreeze456_320.py
```

4. Place test images at ~/freuid/public_test/public_test/ and ~/freuid/private_test/

5. Run inference:

```
python final_inference_fast.py
```

Output: final_submission.csv with 142,818 rows (id, label).

### 6.3 Runtime

Training: ~9 minutes per epoch on GTX 1650, ~5 hours for all 5 folds. Inference: ~25 minutes for 142k images on GTX 1650 (batch size 16, 320px, 5-model ensemble). Expected inference time on A100: under 30 minutes.

### 6.4 Randomness

No explicit random seed was fixed during training. The GTX 1650 runs in standard (non-deterministic) mode. Residual non-determinism from GPU floating-point reductions may cause negligible variation in fraud scores across runs.

### 6.5 Repository

Code: https://github.com/Janet-Priya/freuid-challenge-2026

Commit SHA: e2ec010618b879d32c30c08f394b316503abf456

License: MIT

## 7. Team and Contributions

Solo submission. Janet Anand designed and implemented the full pipeline: data exploration, validation strategy, model architecture, training loop, augmentation, ensemble, inference, and Docker packaging.